HybridSense

An AI Framework That Makes Buildings Think for Themselves - Without Replacing a Single Wire

BACnet/IP | Siemens Desigo CC | Random Forest | ISO 7730 PMV

March 2026 - Progress

The Problem

Why today's buildings waste energy they don't need to

40%

of all building energy goes to HVAC systems

Most of it is avoidable. HVAC runs even when no one is in the room.

Buildings account for ~40% of global energy consumption (IEA 2023)

HVAC Runs on a Clock

Building management systems operate on fixed time schedules set at installation. The system has no way of knowing whether a room is actually occupied, it just follows the calendar.

Empty rooms get fully heated or cooled, all day, every day.

No Easy AI Layer Exists

Smarter control exists in research, but deploying it commercially requires either expensive new sensors (CO_2, PIR, cameras) or a full BMS replacement - neither is realistic at scale.

The gap between 'monitoring data' and 'acting on it' remains wide open.

The Cost Adds Up Fast

In a typical commercial building, 20–30% of HVAC energy is consumed in unoccupied spaces. For a 10,000 m² office, that's tens of thousands of euros in preventable waste per year.

And with every building, the problem multiplies.

The good news: buildings already collect the data needed to solve this. The missing piece is a layer of intelligence that uses it.

THE SOLUTION

HybridSense sits silently above your existing BMS.

It reads sensor data from the building, predicts whether each space is occupied, and adjusts setpoints automatically.

- No new hardware
- Existing BMS unchanged
- Runs every 15 minutes

Fail-safe design: if the AI module loses connection, the building reverts automatically to standard BMS control.

How It Works

A 5-step loop that repeats every 15 minutes, around the clock

Every 15 minutes, HybridSense runs a complete sense-predict-decide-act cycle. Here's exactly what happens:

1 READ

HybridSense connects to the Siemens PXC controller via Ethernet and reads three sensor values (return air temperature, supply humidity, and return duct pressure) using the BACnet/IP protocol.

2 COMPUTE FEATURES

From the raw sensor data, five physical features are calculated: temperature, humidity, humidity change, temperature rate of change, and duct pressure. These carry the physical "fingerprint" of human presence.

3 PREDICT OCCUPANCY

A Random Forest model trained on 8,182 records of real Siemens Desigo CC data predicts whether the space is occupied (1) or empty (0). The model was carefully designed to avoid data leakage.

4 CHECK COMFORT (PMV)

Before any action, the system calculates a thermal comfort index (PMV per ISO 7730). If comfort would be compromised by reducing the setpoint, the action is blocked , regardless of the occupancy prediction.

5 WRITE SETPOINT

If occupied → setpoint stays at 22°C.
If empty and PMV is safe → setpoint is raised to 24°C (2°C setback). The command is written via BACnet to Desigo CC.
Everything is logged to a local database.

The entire cycle read, compute, predict, check, write — takes under 2 seconds. The 15-minute interval matches standard BMS control granularity.

System Architecture

HybridSense adds intelligence at the top , touching nothing below

4 **HybridSense — AI Supervisor**

Occupancy prediction · PMV safety check · Setpoint decision · 15-min loop

3 **BMS — Siemens Desigo CC**

Receives setpoint commands · Validates against safety limits (20–26°C hardcoded)

2 **DDC Controllers (PXC / DXR)**

PID control loops · BACnet/IP network ◀ COMPLETELY UNCHANGED

1 **Field & Actuators**

VAV boxes · Control valves · Fan drives · Temperature / Humidity / Pressure sensors ◀ UNCHANGED

DESIGN PRINCIPLES

Zero hardware

No sensors, cables, or devices to install. Works with what's already there.

Non-invasive

DDC controllers and field devices are never modified or interrupted.

Fail-safe

If AI module fails or disconnects, the building defaults to standard BMS control automatically.

Audit trail

Every decision cycle — prediction, PMV value, setpoint sent — is logged to a local SQLite database.

BACnet/IP
Write Property

Data & Model

How the occupancy prediction was trained — and why it's reliable

Where the Data Comes From

Training data was collected from Siemens Desigo CC at Bilkent Hospital — a controlled, 24/7 BMS environment with rich, clean sensor streams.

The dataset covers 87 days (Nov 2025 – Jan 2026), sampled every 15 minutes, giving 8,182 labelled records after cleaning.

8,182	87	15 min	5
records	days	interval	features

Two methodological problems were identified and resolved during development:

1. Label leakage: HVAC supply signals were used to generate occupancy labels AND as model features. This inflated AUC to 0.998. All supply-side signals were removed. AUC dropped to 0.798 .
2. Domain mismatch : hospital OR ≠ office environment. Features were chosen for physical transferability, not environment-specific behaviour.

The 5 Features / Why These, Not Others

return_temp_C *Return air temp*	People raise ambient T through metabolic heat
supply_humidity_pct *Supply humidity*	Occupants add moisture through respiration
humidity_diff *Humidity change*	Captures arrival/departure transitions
temp_roc *Temp rate of change*	Drift pattern differs: occupied vs empty
return_dp_Pa *Return duct pressure*	Occupancy-driven ventilation demand

All 5 features are return-side / derived signals - they carry the physical trace of occupancy without encoding HVAC control decisions.

Chronological 80/20 split was used (not random shuffle) because the model must predict FUTURE occupancy from PAST data, just like real deployment.

Model Results

Why Random Forest — not the model with the highest score

Performance on Test Set (1,637 records)

Metric	Random Forest	Log. Regression
AUC-ROC	0.832	0.874
F1-Score	0.786 ✓	0.660
Recall	0.736	0.517
Precision	0.843	0.913
Accuracy	0.725	0.635
Specificity	0.701	0.893

The key question: which error is worse?

False Negative (FN) [HIGH risk]

Model says 'empty' — room is actually occupied

HVAC reduces output → temperature drifts → occupants are uncomfortable

False Positive (FP) [LOW risk]

Model says 'occupied' — room is actually empty

HVAC runs unnecessarily → energy wasted, but nobody is affected

Random Forest decision:

RF produces 296 False Negatives vs LR's 542 — that's 46% fewer comfort violations. Higher AUC matters less than protecting the people in the building.

Backup layer: If RF still misclassifies and setback is about to cause discomfort, the PMV constraint catches it automatically.

AUC from roc_analysis.py (n_est=300) · F1/Recall/Precision/Accuracy/Specificity at default threshold (0.5) · PKL deployed models

Thermal Comfort & Energy Results

ISO 7730 compliance maintained across all scenarios — 2°C setback is optimal

What is PMV?

PMV (Predicted Mean Vote) is the ISO 7730 standard index for thermal comfort. It runs from -3 (too cold) to +3 (too hot), with 0 being perfect neutral.

ISO 7730 Category B-used for offices and hospitals requires:

-0.5 ≤ PMV ≤ +0.5
PPD (dissatisfied) < 10%

In HybridSense, PMV serves two roles:

1. Safety constraint: before any setback is applied, PMV is checked. If applying the setback would push PMV below -0.5, the action is blocked.

2. Validation metric: used in WP4 to confirm real-world comfort is maintained.

	Baseline	1°C Setback	2°C Setback	3°C Setback
	Static BMS (no AI)		★ *OPTIMAL*	
PMV	+0.348	+0.294	+0.241	+0.189
PPD	8.0%	7.3%	7.1%	7.3%
In comfort	87.5%	90.8%	90.8%	90.7%
Occ. risk	0 slots	0 slots	0 slots	11 slots ⚠
		Energy: −6.6%	Energy: −16.7%	Energy: −25.1%

The 2°C setback improves mean PMV from +0.348 to +0.241 , meaning thermal comfort actually gets BETTER while energy is saved. Zero occupancy comfort risk.

WP4 — Real-World Validation

April 2026 · Siemens Demo Room · Actual BACnet/IP closed-loop integration

This isn't a simulation. WP4 will connect HybridSense to a live Siemens Desigo CC system and run the full predict-decide-act cycle on real hardware.

1 BACnet/IP Setup

Connect laptop → PXC00.ED controller via Ethernet · Run BAC0 whois to discover devices · Confirm Desigo CC is live

2 Live Sensor Read

Read return_temp_C, supply_humidity_pct, return_dp_Pa as BACnet Analog Inputs · Confirm values streaming to Python

3 Inject Test Scenarios

Siemens engineer injects 5 sensor value sets into Desigo CC representing: occupied, empty/safe, PMV-edge, high temp, threshold edge

4 Full Closed-Loop Run

For each scenario: predict → calculate PMV → decide → write setpoint · Entire cycle logged with timestamps

5 Export & Verify

Export CSV decision log + Desigo CC trend log + screenshots · Cross-check decisions against expected outcomes

Success criteria: BACnet connection established · Read Property ≥99% success · Write Property confirmed in PXC · S3 scenario (PMV constraint) correctly cancels setback · Full 5-scenario log exported

Hardware confirmed: PXC00.ED · PXC7.E400L · PXC4.E16 · DXR2 room controllers · RDG thermostats · TXM1 I/O modules (present in Siemens demo room).

Why This Matters for Sensgreen

The exact capability gap HybridSense closes

From Monitoring to Control — Sensgreen's 2026 Direction

Dimension	Sensgreen Today	+ HybridSense
BMS Integration	BACnet/Modbus/KNX connect → Building X integration	Same stack — I work in it daily at Siemens Smart Infrastructure
Occupancy	Monitoring + analytics	Closed-loop setpoint control (the next layer)
AI Role	Detect anomalies Prioritize actions	Make control decisions autonomously
Energy Impact	Identifies savings	Executes savings via BACnet write
Hardware	LoRaWAN sensor layer	Zero — uses existing BMS signals only

Contributor Profile

Daily work
Siemens Smart Infrastructure
Desigo CC · Building X
BACnet/IP · KNX · Modbus TCP

Academic
Mechatronics + EEE double major
Kadir Has University
Full scholarship · GPA 3.46

HybridSense
Active capstone project
WP1–WP3 complete
WP4 demo: April 2026

Recognition
AWS AI4GOOD Hackathon
Berlin 2026 · 2nd place

Availability
Graduating May 2026
Ready to contribute immediately

Sensgreen's 2026 roadmap: 'Turning recommendations into tested control logic.' HybridSense is a working prototype of exactly that.

Where We Are — and Where This Goes

WP1 Data & ETL

87-day BMS dataset
ETL pipeline built
BACnet point mapping

Nov–Dec 2025

✓ Done

WP2 Architecture

4-layer system design
Algorithm selection
PMV methodology

Jan 2026

✓ Done

WP3 Model & Platform

Python pipeline
PMV module
Streamlit dashboard

Feb–Mar 2026

✓ Done

WP4 Real BMS Demo

Siemens demo room
Closed-loop BACnet/IP
Validation & report

April 2026

→ Active

WP1–3 ✓
WP4 → Apr 2026